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Q1.

(1) 1阶，线性（∵ $y'+y=4x^2$）

(2) 2阶，no（∵ $\left(\frac{dy}{dx}\right)^2$）

(3) 1，no（∵ $(y')^2 \cdot y^2$）

(4) 2，线性（$\sin x$ ⇒ Const.）

(5) 1，no（∵ $\cos y$）

(6) 2，no（∵ $\sin y''$）

Q2 $\frac{d^2y}{dx^2}+\omega^2 y=0$，$\omega>0$，且 $\omega\in\mathbb{R}$

	$\boxed{y}$	$\boxed{y'}$	$\boxed{y''}$	$y''+\omega^2 y$
(1)	$\cos\omega x$	$-\omega\sin\omega x$	$-\omega^2\cos\omega x$	$=0$
(2)	$C_1\cos\omega x$	$-C_1\omega\sin\omega x$	$-C_1\omega^2\cos\omega x$	$=0$
(3)	$\sin\omega x$	$\omega\cos\omega x$	$-\omega^2\sin\omega x$	$=0$
(4)	$C_2\sin\omega x$	$C_2\omega\cos\omega x$	$-C_2\omega^2\sin\omega x$	$=0$
(5)	$C_1\cos\omega x+C_2\sin\omega x$	（∵ (3)+(4)）		$=0$
(6)	$A\sin(\omega x+\beta)$	$A\omega\cos(\omega x+\beta)$	$-A\omega^2\sin(\omega x+\beta)$	$=0$

线性叠加

Q3

	$\boxed{y}$	$\boxed{y'}$	LHS =? RHS
(1)	$\frac{\sin x}{x}$	$\frac{\cos x}{x}-\frac{\sin x}{x^2}$	是
(2)	$2+c\sqrt{1-x^2}$	$\frac{c}{2\sqrt{1-x^2}}(-2x)$	是
(3)	ce^x	ce^x	是
(4)	e^x	e^x	是
(5)	$\sin x$	$\cos x$	是
(6)	$-\frac{1}{x}$	$\frac{1}{x^2}$	是
(7)	x^2+1	$2x$	是
(8)	$\frac{g(x)}{f(x)}$	$\frac{g'(x)}{f(x)}-\frac{g(x)}{f^2(x)}f'(x)$	

题设 $y'=\frac{f(x)}{g(x)}y^2-\frac{f'(x)}{f(x)}$

$=\frac{f(x)}{g(x)}\left(\frac{g(x)}{f(x)}\right)^2-\frac{f'(x)}{f(x)}$

$=\frac{f'g-f'f}{f^2}$ （否）

Q4 $\frac{dy}{dx} = 2x$

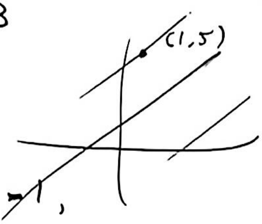
(1) $y = x^2 + C$

(2) $4 = 1^2 + C$, $\therefore C = 3$

特解: $y = x^2 + 3$

(3) $/\!/ l: y = 2x+3$,

$k = 2$, $2x = 2$, $x = 1$,



$y = 2(1) + 3 = 5$,

取 $(1,5)$

$5 = 1^2 + C'$, $\therefore C' = 4$

$y = x^2 + 4$

(4) $\int_0^1 y\,dx = 2$

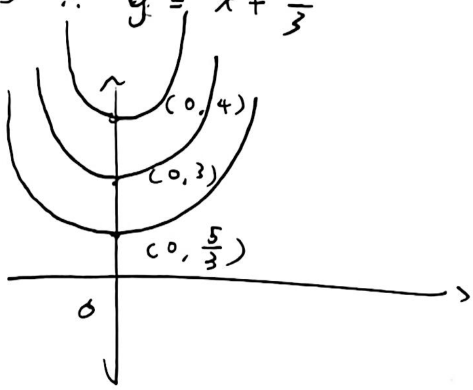
$\because y = x^2 + C$

$\int_0^1 (x^2 + C'')dx = 2$

$\frac{x^3}{3} + C''x \Big|_0^1 = 2$

$(\frac{1}{3} + C'') - (0) = 2$, $\therefore C'' = \frac{5}{3}$

(5) $\therefore y = x^2 + \frac{5}{3}$



l_2: $y = x^2 + 3$

l_3: $y = x^2 + 4$

l_4: $y = x^2 + \frac{5}{3}$

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Q1 (1) $\frac{dy}{dx} = 2xy$, $x = 0, y = 1$

$\ln|y| = x^2 + C$, $0 = 0 + C$, $\therefore C = 0$

$y = e^{x^2}$

(2) $y^2 dx + (x+1)dy = 0$, $(0,1)$

$C + \ln|x+1| = \frac{1}{y}$

$C = 1$, $\therefore y = \frac{1}{1 + \ln|x+1|}$

(3) $\frac{dy}{dx} = \frac{1+y^2}{xy + x^3y}$

$\int \frac{y}{1+y^2}dy = \int \frac{1}{x(1+x^2)}dx = \int (\frac{1}{x} - \frac{x}{1+x^2})dx$

$u = 1 + y^2$

$\frac{1}{2}\ln|1+y^2| = \ln|x| - \frac{1}{2}\ln(1+x^2) + C$

$\cancel{1 + y^2 = \frac{Cx}{1+x^2}}$

$1 + y^2 = \frac{C''x^2}{1+x^2}$, $C'' = e^{C'} = e^{(2C)}$

(4) $(1+x)y\,dx + (1-y)x\,dy = 0$

$\int \frac{1+x}{x}dx + \int \frac{1-y}{y}dy = 0$

$\ln|x| + x + \ln|y| - y = C$

$\cancel{\ln|xy| = C - x + y}$

$e^{\ln|xy| + x - y} = e^C$

$\ln|xy| \cdot e^{x-y} = e^C = K$

$|xy| e^{x-y} = K$

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Q1 (5) $\cancel{(y+x)dy + (x-y)dx = 0}$

$\cancel{\frac{dy}{dx} = \frac{y-x}{y+x}}$

$\cancel{= \frac{1-\frac{x}{y}}{1+\frac{x}{y}}}$

$\cancel{u = \frac{x}{y},\quad x = yu}$

$\cancel{\frac{dx}{dy} = u +}$

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(5) $(y+x)dy + (x-y)dx = 0$

$$\frac{dy}{dx} = \frac{y-x}{y+x}$$

$$= \frac{\frac{y}{x}-1}{\frac{y}{x}+1}$$

令 $y = ux$, $\frac{dy}{dx} = u + x\frac{du}{dx}$

$$\therefore \frac{dy}{dx} = u + \frac{du}{dx} = \frac{u-1}{u+1}$$

$$\frac{du}{dx} = -\frac{1+u^2}{u+1}$$

$$\int \frac{u+1}{1+u^2}du = -\int \frac{1}{x}dx$$

$$\frac{1}{2}\ln(1+u^2) + \arctan u = -\ln|x| + C$$

代回 $u = \frac{y}{x}$,

$$\therefore \ln\sqrt{x^2+y^2} = \arctan(\frac{y}{x}) = C$$

Q1 (6)

$$x\frac{dy}{dx} - y + \sqrt{x^2+y^2} = 0$$

$$\frac{dy}{dx} = \frac{y}{x} - \sqrt{1-(\frac{y}{x})^2}$$

~~u = yx~~, 令 $y = \frac{u}{x}$

$$u + x\frac{du}{dx} = u - \sqrt{1-u^2}$$

$$\int \frac{1}{\sqrt{1-u^2}}du = -\int \frac{1}{x}dx$$

$$\arcsin\frac{y}{x} + \ln|x| = C$$

注: $1-u^2=0$, 即 $y = \pm x$ 也为原方程解

(7) $\tan y\, dx = \cot x\, dy$

$$\int \tan x\, dx = \int \cot y\, dy$$

$$-\ln|\cos x| = \ln|\sin y| + C_1$$

$$\ln|\sin y \cos x| = C_2$$

$$\therefore \sin y \cos x = C$$

(8)

$$\frac{dy}{dx} = -\frac{e^{y^2}\cdot e^{3x}}{y}$$

$$\int \frac{y}{e^{y^2}}dy = \int -e^{3x}dx$$

$u = -y^2$, $-2y\,dy = du$

$$3e^{-y^2} - 2e^{-3x} = C$$

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Q1 (9) $\frac{dy}{dx} = \frac{y/x}{-\ln(y/x)}$

$u = y/x$, $y = ux$, $\frac{dy}{dx} = u + x\frac{du}{dx}$

$u + x\frac{du}{dx} = \frac{u}{-\ln u}$

$\int \frac{\ln u}{u(1+\ln u)} du = -\int \frac{1}{x} dx$

$v = 1+\ln u$, $dv = \frac{1}{u} du$

$\ln u - \ln|1+\ln u| = -\ln|x| + C$

$\therefore \frac{y}{1+\ln(\frac{y}{x})} = C$

(10) $\frac{dy}{dx} = e^{x-y}$

$\int \frac{1}{e^{-y}} dy = \int e^x dx$

$e^y = e^x + C$

Q2 (1) $\frac{dy}{dx} = \frac{1}{(x+y)^2}$

$u = x+y$. $\frac{du}{dx} = 1 + \frac{dy}{dx}$

$\frac{du}{dx} = 1 + \frac{1}{u^2} = \frac{u^2+1}{u^2}$

$\int \frac{u^2}{u^2+1} du = \int dx$

$y - \arctan(x+y) = C$

Q2

(3) $\frac{dy}{dx} = \frac{x-y+5}{x-y+2}$

$u = x-y$. $\frac{du}{dx} = 1 - \frac{dy}{dx}$

$\frac{du}{dx} = 1 - \frac{u+5}{u-2} = -\frac{7}{u-2}$

$\int (u-2) du = \int -7 dx$

$(x-y)^2 + 10x + 4y = C$

(4) $\frac{dy}{dx} = (x+y+1)^2$

$u = x+y+1$. $\frac{du}{dx} = 1 + \frac{dy}{dx} = 1 + u^2$

$\int \frac{1}{1+u^2} du = \int dx$

$\arctan(x+y+1) = x + C$

(5) $\frac{dy}{dx} = \frac{y^6 - 2x^2}{2xy^5 + x^2y^2}$

$y^3 = z$, $3y^2 \frac{dy}{dx} = \frac{dz}{dx}$

$\frac{dz}{dx} = \frac{3(z^2 - 2x^2)}{2xz + x^2}$

$z = ux$, $\Rightarrow (u-3)^7 (u+2)^3 = Cx^{15}$

$\therefore (y^3 - 3x)^7 (y^3 + 2x)^3 = Cx^{15}$

~~(2) $\frac{dy}{dx} = \frac{2x-y+1}{x-2y+1}$~~

$\begin{cases} 2x - y + 1 = 0 \\ x - 2y + 1 = 0 \end{cases}$, 交点 $(-\frac{1}{3}, +\frac{1}{3})$

$x = X - \frac{1}{3}$, $y = Y + \frac{1}{3}$

齐次化. $\frac{dY}{dX} = \frac{2X - Y}{2Y - X}$. $Y = uX$

$X\frac{du}{dX} = \frac{2(1-u+u^2)}{1-2u}$, $\Rightarrow$

$\int \frac{1-2u}{2(u^2-u+1)} du = \int \frac{1}{X} dX$

$x^2 - xy + y^2 + x - y = C$

Q2 (6) $\frac{dy}{dx}=\frac{2x^3+3xy^2+x}{3x^2y+3y^3-y}$

解 $u=x^2$，$v=y^2$，

$$\frac{dv}{du}=\frac{y}{x}\left(\frac{dy}{dx}\right)=\frac{2x^2+3y^2+1}{3x^2+2y^2-1}=\frac{2u+3v+1}{3u+2v-1}$$

方程交点 $(1,-1)$，齐次变化求解

$$(y^2-x^2+2)^5=C(x^2+y^2)$$

Q3 $\frac{x}{y}\frac{dy}{dx}=f(xy)$，$xy=u$，$\frac{du}{dx}=y+x\frac{dy}{dx}$，$x\frac{dy}{dx}=\frac{du}{dx}-\frac{u}{x}$

~~(1) $y(1+xy^2)dx=x\,dy$~~

$$\frac{1}{u/x}\left(\frac{du}{dx}-\frac{u}{x}\right)=f(u)$$

$$\frac{x}{u}\frac{du}{dx}-1=f(u)$$

$\therefore$
$$\frac{1}{u(f(u)+1)}du=\frac{1}{x}dx$$

(1) $\frac{dy}{dx}=\frac{y(1+x^2y^2)}{x}$

$\Rightarrow$ $\frac{x}{y}\frac{dy}{dx}=1+x^2y^2$

$f(u)=1+u^2$，代入

$$\frac{1}{2}\ln|u|-\frac{1}{4}\ln|u^2+2|=\ln|x|+C$$

$$\frac{y^2}{x^2y^2+2}=Cx^2$$

2) $\frac{x}{y}\frac{dy}{dx}=\frac{2+x^2y^2}{2-x^2y^2}$

$f(u)=\frac{2+u^2}{2-u^2}$

$\frac{x}{u}\frac{du}{dx}=\frac{4}{2-u^2}$ $\Rightarrow$ $\frac{2-u^2}{4u}du=\frac{1}{x}dx$

$$\ln\left|\frac{y}{x}\right|-\frac{x^2y^2}{4}=C$$

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Q5 $x(t+s) = \dfrac{x(t)+x(s)}{1-x(t)x(s)}$，$x'(0)$ 存在

(I) 令 $t = s = 0$，$x(0) = \dfrac{2x(0)}{1-x^2(0)}$

$x(0) - x^3(0) = 2x(0)$，$\therefore x(0) = 0$

(II) $x'(t) = \lim\limits_{s\to 0} \dfrac{x(t+s)-x(t)}{s} = \lim\limits_{s\to 0} \dfrac{\dfrac{x(t)+x(s)}{1-x(t)x(s)} - x(t)}{s}$

$x'(t) = \lim\limits_{s\to 0} \dfrac{x(t)+x(s) - x(t) + x^2(t)x(s)}{s[1-x(t)x(s)]}$.

(III) $\because x(0) = 0$，$\lim\limits_{s\to 0} \dfrac{x(s)}{s} = x'(0) = k \in \mathbb{R}$

$\therefore \quad x'(t) = k(1+x^2(t))$

(IV) $\dfrac{1}{1+x^2}dx = k\,dt \Rightarrow \arctan x = kt + C$

(V) $x(t) = \tan(kt)$，其中 k 由 $x'(0)$ 确定常数